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**CLASS :** **BSCS-E**

**CMS-ID :** **588984**

**ASSIGNMENT :** **01**

**COURSE :** **Applications of ICT**

**COURSE INSTRUCTOR :** **MS. Kashaf Rahim**

**Step 1: Find a Real Problem (30–45 min)**

Walk around your campus, hostel, local market, or home neighbourhood. Identify one specific, small-scale problem that a piece of technology could realistically help with. Not "healthcare in Pakistan" but something like "the canteen line at 1pm always backs up because orders are taken by hand" or "my grandmother can't tell which of her medicines to take when."

Deliverable: 3–5 sentences describing the problem, where you observed it, and who is affected.

## PROBLEM:

At GATE NO.2 of NUST, I observed that heavy traffic builds up during rush hours.

The security guards stop every vehicle and check the student cards and collect the CNIC of parents / drivers. This creates traffic at the gate, and because of this many students arrive late to class in the morning. This problem is affecting their studies and causing frustration. This system wastes around 10-15 minutes of every student's time, who comes by car. So, this problem must be addressed.

**Step 2: Brainstorm, on Paper First (45–60 min)**

Before touching any AI tool, generate at least 8 possible ICT-based solutions to your problem using one of these techniques (pick one):

- Mind mapping
- "How Might We..." question chains

This must be done by hand, photograph or scan your actual notes/sketches. Messy is fine; this is evidence of your thinking, not a final product.

Deliverable: Photo/scan of handwritten brainstorming + a typed list of your 8+ ideas (one line each).

## BRAINSTORM ON PAPER:

(Students who forget their cards at home should not be stopped at the gate us it to disturb the traffic instead send them to side checking area. (without technology)

(There must be 2 lanes (different) for regular and visitors. to keep them separate and will be easy to manage).

(Students suggestions must be taken to improve the system).

(Process)

(Deploy IOT based connected system in the whole process)

(This approach may reduce the chances of misiakes and improve accuracy).

(Smart scanners, smart databases to store data & processing)

Gate NO2; (Show ID & CNIC Check)

(People)

(Student, Guards, parents)

(Proper training sessions should be arranged for the guards to guide how to use this auto-maied system).

(Techology)

(Camera detections for regular cars and let them pass without disturbance to maintain traffic flow)

(Automated CNIC card. card scanners to remove token systems)

(Clear instructions boards for the scanning of both things so that new drivers should know ~~which~~ what is required to get entry).

(Time/Traffic)

(Apply smart technologies like IOT systems in the whole procedure to maintain Traffic flow).

(Student cards must have a unique code which when scanned. give authentication about his/her data).

## TYPED LIST OF IDEAS:

**1)** Two separate lanes for regular vehicles and visitors

**2)** A side checking area (no technology) for students who forget their cards, so traffic is not blocked

**3)** Camera detection for regular cars so they can pass without stopping

**4)** Automated CNIC and card scanners to replace the token system

**5)** A unique code on student cards that gives authentication when scanned

**6)** An IOT- based connected system for the whole entry process

**7)** Clear instructions boards for new drivers about what is required for entry

**8)** Training sessions for guards on how to use the automated system

**9)** Taking student's suggestions to improve the system

**Step 3: Narrow and Defend (30 min)**

From your 8+ ideas, pick your top choice and explain why, using a simple comparison table (see below, fill in with your own ideas):

| Idea | Feasible with student budget? | Solves the real problem? | Someone would actually use it? |
|---|---|---|---|
| **1.Two separate lanes** | **Yes, it is low cost (mainly needs road marking and signboards)** | **Yes, it can reduce traffic by keeping regular cars and visitors in separate lanes** | **Yes, students and visitors can easily understand and use the lanes** |
| **2.Side checking area** | **Partly (needs some extra space and a guard to handle students separately)** | **Yes, students who forget their cards can be checked on the side without stopping the main traffic** | **Yes, students who forget their cards can use this area instead of blocking the main lane** |
| **3.Automated CNIC and card scanners** | **No, it can be expensive (buying and installing scanners can be expensive for a student level)** | **Yes, it can make checking faster and reduce waiting time** | **Yes, students and guards would use it if system is easy to operate** |

## TOP CHOICE:

My top choice is two separate lanes.

The reason is that it is low cost and can help reduce traffic at the gate. Regular vehicles and visitors can use different lanes, so the checking process can be faster. The side checking area needs extra space and guards, while automated scanners can be expensive. However, lanes alone do not remove the CNIC checking time.

**Step 4: Ask GenAI, Then Argue with It (45–60 min)**

Now use any GenAI tool (ChatGPT, Claude, Gemini, etc.) to get a second opinion on your chosen idea. Prompt it to critique or expand your solution.

## EXACT PROMPT USED: (CLAUDE)

I am a first-semester BSCS student. At Gate 2 of my university security guards stop every vehicle, check the student card and collect the driver's CNIC. This causes long queues in the morning rush hours and students arrive late to class. My chosen solution is to create two separate lanes one for regular vehicles (students with cards) and one for visitors. Please critique this idea honestly. Tell me its weaknesses, what could go wrong, and how I could improve it using low -cost ideas suitable for students. Keep your answer short and simple.

## SUMMARY OF AI SUGGESTIONS:

AI said that two separate lanes are a good and low  cost idea, but they will not make the checking process faster by themselves. There may not be enough space at the gate, and some drivers may also use the wrong lane. Students who forget their cards can still cause a blockage. Clear signboards can help and regular vehicles could have stickers so guards can identify them quickly. A separate area for students without cards and one extra guard during rush hours could also help. Overall, separate lanes are a good first step, but they should be used with a faster checking method.

## DISAGREEMENT WITH AI:

**1)** The AI suggested using physical stickers on regular vehicles for quick identification at the gate. I disagree with this because managing physical stickers for hundreds of student vehicles involves extra manual work (printing, distribution etc). Stickers can also be easily damaged or lost.  Instead of this a digital QR code linked to the student's portal would be a much safer and faster solution.

**2)** The AI suggested creating a separate checking area with extra guards for students who forget their cards. As I evaluated in my comparison table, adding dedicated guards

and extra space increases operational costs and is not very practical. Instead of adding manual staff, we can simply integrate a quick digital system (such as a mobile app or an automated scanner) at the main gate. So, guards can verify cardless students digitally without increasing the labour.

**Step 5: Final Write-Up (30–45 min)**

5. One paragraph: what would you need (skills, tools, people) to build this?

To build this solution, I would need skills such as **BASIC TRAFFIC FLOW MANAGEMENT, PLANNING LANE WIDTHS, SIMPLE PROJECT MANAGEMENT, and TEAM COORDINATION**. The tools required would be **ROAD PAINT, TRAFFIC CONES, CLEAR DIRECTIONAL SIGN BOARDS, QR CODE SCANNER, and VERIFICATION PORTAL OF STUDENT CARDS.** I would also need help from people like **UNIVERSITY MANAGEMENT, COORDINATION OF CAMPUS SECURITY TEAM, and ASSISTANCE FROM IT DEPARTMENT**. With these things, the lanes can be set up at low to medium cost.